# Listen Before You Classify: An Adaptive DSP Wake-Up Front End for Energy-Efficient Emergency Sound Alerts on Wearables

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***Abstract*—[written last, from the results]**

***Index Terms*—sound event detection, wake-up detector, Goertzel algorithm, noise-floor tracking, assistive technology, energy efficiency**

## I. Introduction

Wearable sound-awareness devices turn important environmental sounds, such as smoke alarms, sirens and door knocks, into vibration and visual alerts for Deaf and hard-of-hearing (DHH) people [1]. They typically run a pretrained neural classifier such as YAMNet [2] on every half second of microphone audio. This has two costs. First, *energy*: the classifier runs continuously although nothing of interest happens most of the time, and battery life was the main deployment constraint reported for on-watch sound recognition [1]. Second, *safety*: general-purpose classifiers are trained on everyday audio and can be weak on standardized emergency signals, such as the ISO 8201 temporal-three (T-3) smoke-alarm pattern [3].

Both problems have a classical signal-processing answer. Emergency alarms are *designed* signals: a known pitch and a known on/off rhythm, which a matched detector can recognise directly. And most of the day nothing new happens, which a cheap onset detector can tell without a neural network. We therefore ask: *can an adaptive DSP front end, which detects emergency alarms itself and wakes the classifier only when a new sound starts, cut classifier computation substantially while raising alarm recall, in acoustic environments it was never tuned on?*

Our contributions are: (i) a front end built only from classical DSP blocks (a Butterworth high-pass filter, a short-time spectrum with per-band minimum-tracking noise floor, and a Goertzel resonator bank with a T-3 pattern matcher) that costs about 0.5 % of one classifier inference; (ii) a controlled comparison against continuous inference, naive duty cycling and a fixed energy gate, calibrated on 11 environments and tested on 6 unseen ones; (iii) an open, one-command reproducible benchmark built only from public data.

## II. Related Work

**Sound awareness for DHH users.** SoundWatch [1] classified environmental sounds on a smartwatch and compared watch-only, phone and cloud architectures, finding battery drain a key concern. ProtoSound [4] personalised recognition from few examples. Neither reduces *when* the classifier runs.

**Wake-up detectors and cascades.** Always-on systems commonly place a low-power detector before a costly stage [5]; keyword spotting uses small first-stage models that wake larger ones [6]. Recent work cascades a linear screener with a trimmed YAMNet for violence detection [7]. We instead use interpretable, training-free DSP gates, adapt their threshold to the room through noise-floor tracking, and add a second, *decision-making* DSP path for standardized alarms.

**Compute versus accuracy.** Sigtia et al. compared environmental sound classifiers by accuracy and computational cost, including smoke alarms recorded in homes [8].

## III. Method

### A. Signals and sampling

Audio is sampled at $f_s = 16\,\text{kHz}$, the classifier's native rate. The Nyquist frequency of 8 kHz covers the piezo smoke-alarm band (2.8–3.6 kHz) and the 520 Hz low-frequency alarm required for sleeping areas in some codes [9], together with its odd harmonics up to the 13th (6.76 kHz). Synthetic square-wave alarms are band-limited below 7.5 kHz so that no harmonic aliases. The classifier sees a 0.975 s window (15 600 samples) every 0.5 s.

### B. Classifier and decision rule

YAMNet (MobileNetV1, 521 AudioSet classes) costs $M_Y = 70\,744\,064$ multiply-accumulates (MACs) per window, counted from its convolution and dense layers. Its scores are mapped to three alert categories (fire alarm, siren, door) by taking the maximum over category labels. An alert fires when the category score is at least 0.30 in $N$ consecutive classified windows ($N = 2$ for fire alarm and siren, 1 for door), with a 20 s cooldown per category and the most urgent category winning. A window the front end skips is not classified and breaks a run. The classifier and rule are fixed throughout; only the front end changes.

### C. Gate path

**High-pass filter.** Specification: Butterworth, order 2, $f_c = 100$ Hz ($-3$ dB), maximally flat passband; implemented as one biquad,

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}, \tag{1}$$

with a double zero at $z = 1$ (DC) and two poles inside the unit circle (Fig. **??**). It removes DC offset, handling rumble and wind below the lowest target component (520 Hz) so they cannot wake the classifier.

**Band energies.** Each 20 ms frame ($N = 320$, Hann window) is transformed with a 512-point FFT and the power is summed in $B = 16$ log-spaced bands from 100 Hz to 7.6 kHz. Band powers are smoothed with a one-pole low-pass $\bar{P}_b[t] = (1-\alpha)P_b[t] + \alpha\bar{P}_b[t-1]$, $\alpha = e^{-T/\tau}$, $\tau = 60$ ms, which reduces the variance of the few-bin low bands. Let $E_b[t] = 10\log_{10}\bar{P}_b[t]$.

**Noise-floor tracking.** Each band's floor follows the minimum of $E_b$ with a bounded rise,

$$F_b[t] = \min\big(E_b[t],\, F_b[t-1] + \rho T\big), \qquad \rho = 3 \text{ dB/s}, \tag{2}$$

a simplified form of minimum statistics [10]. It drops to a quieter background at once and absorbs a steady sound (a fan, a running tap) within seconds.

**Novelty and wake-up.** The novelty is $\nu[t] = \max_b(E_b[t] - F_b[t])$, the largest rise of any band above its own floor, so the threshold is relative to the room rather than absolute. Before each classifier window the gate takes the maximum of $\nu$ over the 0.5 s just received; if it exceeds $\Delta$, the classifier runs and stays awake for a hangover $H$. Because the window spans the last 0.975 s, the onset that woke the gate is inside the first classified window: waking adds no delay.

### D. Alarm path

**Goertzel resonator bank.** The power of DFT bin $k$ in a frame is computed by the Goertzel recursion [11]

$$s[n] = x[n] + 2\cos(\omega_k)s[n-1] - s[n-2], \quad |X_k|^2 = s_1^2 + s_2^2 - 2\cos(\omega_k)s_1 s_2, \tag{3}$$

a second-order IIR resonator with poles at $e^{\pm j\omega_k}$ on the unit circle, read once after $N$ samples. It needs one real multiply per sample per probe, against $N\log_2 N$ for a full FFT, which pays off when only a few frequencies matter. Probes are fixed from the standards, not tuned on data: 17 probes from 2.8 to 3.6 kHz (piezo) and 520, 1560 and 2600 Hz (square-wave fundamental and odd harmonics).

**Tonal track and T-3 matcher.** A frame is "on" when the strongest piezo probe holds over 25 % of the frame energy (or the three square-wave probes over 35 %), and the probe energy is within 8 dB of its 1.5 s running peak, which rejects reverberant tails. After closing 100 ms dropouts, a fire alarm is declared when three pulses of 0.3–0.8 s are separated by gaps of 0.3–0.8 s, the ISO 8201 T-3 timing with tolerance. This detector raises the alert by itself, merged with the classifier's alerts under the same cooldown.

### E. Front ends compared

F0 runs the classifier on every window. F1 runs it on every $k$-th window. F2 wakes it when the high-passed frame level exceeds a fixed $T$ dBFS. F3 is the adaptive novelty gate. F4, the proposed system, is F3 plus the alarm path. F0+A (continuous classifier plus alarm path) separates the two effects.

### F. Cost

Per 0.5 s hop the gate costs about [gate ops] MACs and the alarm path [alarm ops] MACs, counted per stage in the code, i.e. [%] of one classifier inference. Compute relative to F0 is $(d\,M_Y + M_{\text{gate}} + M_{\text{alarm}})/M_Y$, where $d$ is the classifier duty, the fraction of windows classified.

## IV. Experimental Setup

**Streams.** Following the event-mixing protocol of DCASE 2017 Task 2 [12], we build 5-minute streams from DEMAND [13] recordings (8 microphones per environment used as separate streams). Each event stream holds 6 targets (2 per category, one per 50 s slot) at an SNR drawn from $\{-5, 0, 5, 10, 20\}$ dB over the event's active frames, and 6 distractors at 0–20 dB that never overlap a target. Each event stream has an *idle* twin with the background only. The background level varies per stream between $-50$ and $-35$ dBFS.

**Events.** Sirens, doorbells and knocks come from FSD50K [14] and ESC-50 [15] (clips with exactly one target label; official splits). Because no public set of real smoke-alarm recordings exists, fire alarms are synthesized to ISO 8201 with jittered timing and pitch, slight FM and a random room impulse response ($RT_{60}$ 0.3–0.8 s): half piezo (2.9–3.5 kHz), half 520 Hz square wave. Distractors are other FSD50K/ESC-50 sounds, half of them "hard" (bells, phones, doors, impacts, beeps, speech, music); clips labelled *Alarm* without a target label are excluded as ambiguous.

**Splits.** Thresholds are chosen on 11 DEMAND environments with tuning clips and alarm seeds; all results are on 6 environments never seen in tuning (one per DEMAND category: living room, office, cafeteria, street, bus, park) with test clips and seeds.

**Calibration.** For each gate, we choose the lowest-duty setting ($k$; $T$ and $H$; $\Delta$ and $H$) whose *gate-caused* miss rate on tuning data, the share of events F0 detects that the gated system misses, is at most 1 %.

**Metrics.** A target is detected if an alert of its category fires between its onset and 1 s after its end. We report recall with 95 % bootstrap intervals, false alarms per hour on idle and event streams, median latency, classifier duty and relative compute. Paired differences in recall are tested with the exact McNemar test.

## V. Results

[from results.json]

## VI. Discussion

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## VII. Conclusion

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## References


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